

1. Leakage-controlled, target-centric datasets

Established datasets (Sequence-based)

Davis

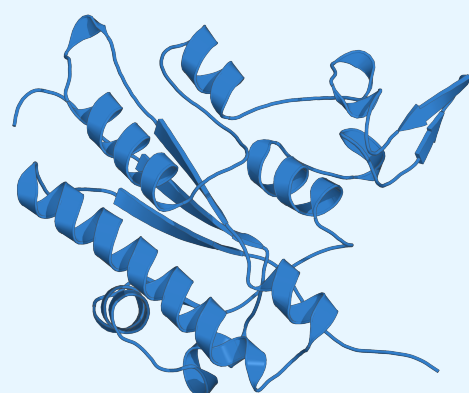
KIBA

Temporal holdout test sets (Structure-based evaluation)

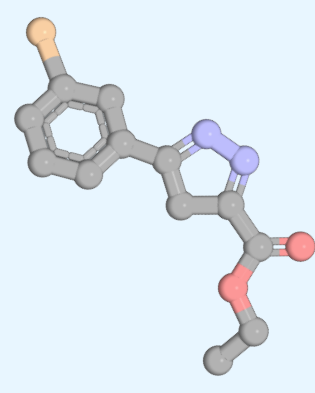
CASP16

ChEMBL35

- No overlap with training data of evaluated models
- Target-centric evaluation (multiple ligands per target)
- AlphaFold3-predicted structures provided for testing



Protein target



Ligand (SMILES)



AlphaFold3
predicted structures

2. Nine deep learning PLA prediction models

Sequence-based models (3)

DeepDTA

LLF

MixingDTA

Structure-based models (6)

FlowDock

Boltz2

FLOWR.ROOT

GraphRG

MFE

LCDD-team

Input

Protein sequence

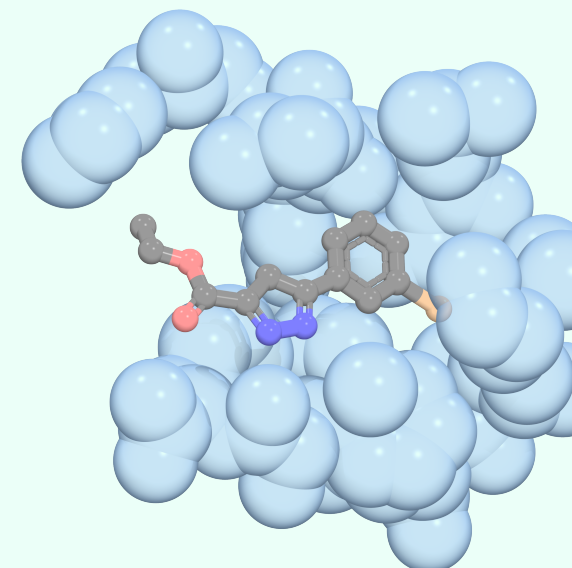
WTATSGSCKGRCFELQE...

Ligand SMILES

CCOC(=O)C1=NN=C(c2c...

Input

Protein-ligand complex structure

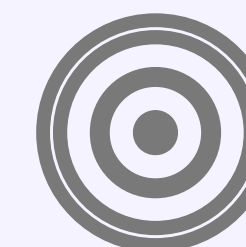


3. Unified evaluation and analysis



Prediction metrics

MSE, RMSE,
Pearson, Spearman, R_m^2 ,
CI, Kendall's τ



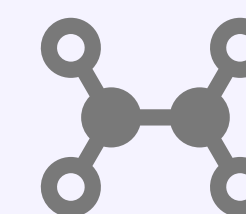
Target-centric evaluation

Per-target and per-family
performance



Structural analysis

Effect of structural accuracy
and molecular interactions



Model complementarity

Cross-model comparison
and ensemble analysis

Key findings from PLABench

1

Large-scale pretrained models achieve the strongest overall performance but do not ensure cross-target generalization

State-of-the-art structure-based models perform best overall, yet sequence-based models show their robustness on some target families.

2

More accurate structures do not consistently improve affinity prediction

Affinity prediction performance does not monotonically improve as protein-ligand structures become more accurate, despite increased recovery of native interactions.

3

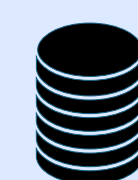
Models exhibit target-dependent blind spots, but show complementary strengths

Performance varies substantially across protein families, with systematic failures for certain target classes. Sequence- and structure-based models provide complementary strengths

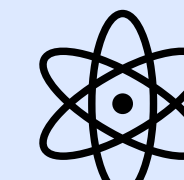
PLABench: An open and extensible framework for rigorous evaluation and development of next-generation protein-ligand binding affinity prediction models



Code



Datasets



AF3 structures



Evaluation pipeline